Tertiary Infotech Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Agent Security

Course Code: C28

Conducted by

Tertiary Infotech Pte Ltd

UEN: 201200696W

Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	30 September 2026	First release of the one-day AI Agent Security course (C28), 9:30am – 5:30pm, 7.5 instructional hours: Topics 1–3 and Activities 1–8.	Dr Alfred Ang

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1. Course Overview

AI Agent Security is a one-day, 7.5-hour course that introduces generative AI, agentic AI and autonomous AI agents, and the security, governance and ethical risks they bring to customer-service and hospitality settings. Learners work entirely on ready-made websites, chatbots and AI agents — no coding is required. The day is organised as three topics mapped one-to-one onto the three learning outcomes.

Topic 1 (LO1) explains how generative AI works and how it becomes agentic — the agentic loop, harness engineering, AI agents such as OpenClaw and Hermes, skills, tools and multi-agent systems. Topic 2 (LO2) covers prompt engineering and post-training, using the Hermes agent on a MiniMax model to analyse data and build presentations. Topic 3 (LO3) covers AI data governance, job impact and redesign, and the cybersecurity risks of autonomous agents, including a safe roll-out framework. Real, dated examples and clearly labelled simulations are grounded in current governance references such as the IMDA Model AI Governance Framework, Singapore's PDPA and PDPC guidance, and the OWASP Top 10 for LLM and Agentic Applications.

Course Details

Element	Detail
Course Code	C28
Duration	1 Day · 7.5 Hours (9:30am – 5:30pm)
Delivery	Instructor-led classroom with hands-on, no-code activities

Learning Outcomes

LO	Learning Outcome
LO1	Demonstrate generative AI concepts and applications relevant to customer service and hospitality management.
LO2	Apply prompt engineering techniques and analyse output variations to improve generative AI performance in service settings.
LO3	Identify ethical risks and analyse bias in AI-generated content used in customer engagement.

2. Course Schedule

The training day runs 9:30am – 5:30pm and delivers 7.5 instructional hours. A 30-minute lunch break is excluded from instructional time; refreshments are taken during activity time.

One-Day Schedule (Topics 1–3)

Slides 1–160 of the trainer deck. The Slides column below gives the deck range for each session.

Time	Topic / Activity	Duration	Slides
9:30 – 9:50	Welcome, introductions, course outline and ground rules	20 min	1–7

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9:50 – 10:35	Topic 1: Brief history of AI 2023-2026; how generative AI works — autoregressive LLM, training and inference	45 min	8–22
10:35 – 11:15	Topic 1: Real generative-AI use cases (AI fashion models, AI video, Klarna); context engineering	40 min	23–34
11:15 – 12:00	Topic 1: Agentic loop and harness engineering; AI agents, OpenClaw and Hermes, skills, tools and multi-agent systems	45 min	35–66
12:00 – 12:45	Activity 1: Talk to the TIA Support AI agent on WhatsApp and post reflections to the Pinboard; debrief	45 min	71–73
12:45 – 1:15	Lunch break	30 min	—
1:15 – 1:30	Topic 1 summary: generative AI vs agentic AI vs AI agents	15 min	67–70
1:30 – 2:05	Topic 2: Install the Hermes agent on MiniMax M2.7; principles of prompt engineering, good vs bad prompts	35 min	74–86
2:05 – 2:55	Activity 2 & 3: Upload mock Excel data and prompt the agent into charts and a .docx insights report, then build a 10-slide PPT and redesign it from an uploaded image template	50 min	87–90
2:55 – 3:20	Activity 4: Install tools and skills, re-run the Excel and PPT tasks; Topic 2 reflection and debrief	25 min	91–94
3:20 – 4:05	Topic 3: AI data governance and accountability (Moffatt v Air Canada); Activity 5 — generate an AI data governance policy with the policy-generator website	45 min	95–109
4:05 – 4:45	Topic 3: Job impact and redesign — why now (12,000+ agents on AgentSea), the stakes (WEF 2030 projections), the human-owner operating model, the problem of adoption debt, the Human Owner role and the GROW intervention; Activity 6 — coaching role-play simulator	40 min	110–122
4:45 – 5:30	Topic 3: AI-agent cybersecurity risks, advanced threats (prompt injection, jailbreaking, data poisoning, supply-chain and agent-to-agent attacks) and safe rollout; Activities 7 & 8 — leaky vs guarded chatbots and security reflection; Topic 3 recap, the business challenge, course feedback and close	45 min	123–160

Instructional total: 450 minutes (7.5 hours), excluding the 30-minute lunch break.

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3. Activities

Every activity uses a ready-made website, chatbot or AI agent — learners do not write any code. Each has its own folder under activities/ containing learner instructions, prompt cards, mock data and printable PDFs. All data is fictional. Full step-by-step facilitation detail is in the Learner Guide.

#	Activity	Topic	Duration	LO
1	Talk to an AI Agent (TIA Support WhatsApp) and reflect on the Pinboard	1	45 min	LO1
2	Analyse mock Excel marketing data into a charted .docx report with the Hermes + MiniMax agent	2	25 min	LO2
3	Create a 10-slide PPT with the agent, then redesign it from an uploaded image template	2	25 min	LO2
4	Install tools and skills, then re-run the Excel and PPT tasks	2	25 min	LO2
5	Generate an AI Data Governance Policy with the policy-generator website	3	30 min	LO3
6	Coach a worried team member (AI role-play simulator)	3	30 min	LO3
7	Leaky vs guarded chatbot — see and stop a PII leak	3	30 min	LO3
8	Reflect on agent security and decide go / no-go	3	15 min	LO3

Times overlap with the topic teaching blocks in the schedule above; activity folders and the reflection Pinboard (alfredang.github.io/pinboard) are used throughout the day.

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4. Tools and Resources

Resource	Purpose
Trainer slide deck (PPTX/PDF)	Facilitation, diagrams and concept walkthroughs
Learner Guide	Detailed notes and step-by-step activity walkthroughs
Activity packs (activities/)	Ready-made websites/chatbots, prompt cards and mock data
TIA Support on WhatsApp (+65 8866 6375)	Live OpenClaw-powered AI agent for Activity 1
Reflection Pinboard	alfredang.github.io/pinboard — group reflections by risk theme
Hermes Agent + MiniMax	hermes-agent.nousresearch.com and minimax.io for Topic 2
LMS/TMS portal	lms-tms.tertiaryinfotech.com — course material and activity files
IMDA / PDPC / OWASP references	Governance and security reference material

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